Beyond Functional Correctness

Towards a Multidimensional Composite Quality Index for AI-Generated Codebases

Research Proposal

SCS 3216 : Research Methods

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Declaration

The literature review is my original work and has not been submitted previously for any examination/evaluation at this or any other university/institute. To the best of my knowledge, it does not contain any material published or written by another person, except as acknowledged in the text.

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This is to certify that this literature review is based on the work of Mr S.M.Silva under my supervision. The literature review has been prepared according to the format stipulated and is of acceptable standard.

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List of Acronyms

BLEU Bilingual Evaluation Understudy

GPT Generative Pretrained Transformer

LLM Large Language Model

Contents

List of Acronyms **ii**

1 Introduction **1**

1.1 Domain 2

1.2 Motivation 2

2 Problem Statement and Justification **4**

2.1 The Research Gap 4

2.2 Justification 5

2.2.1 The Efficiency Paradox: The Trade-off of Sustainability . . . 5

2.2.2 The Success/Maintainability Paradox: The "Add it and Forget it" Anti-pattern 5

2.2.3 The Security-Functionality Disconnect: The Alignment Tax 6

3 Research Aims, Questions, and Objectives **7**

3.1 Primary Aim 7

3.2 Research Questions 7

3.3 Research Objectives 8

4 Conclusion **9**

Chapter 1

Introduction

The software engineering landscape is currently undergoing a paradigm shift, transitioning from traditional human-centric development to a model of Artificial Intelligence-Human Paired Programming. This transformation is driven by the rapid integration of Large Language Models (LLMs) such as GPT-4, Claude, and GitHub Copilot into the Software Development Life Cycle (SDLC). Empirical evidence suggests that these tools significantly enhance developer productivity, with studies reporting speed increases of up to 56% and substantial improvements in throughput. However, this surge in velocity has introduced a "Success Paradox": models that achieve high scores in functional correctness—the ability to pass predefined unit tests—often produce codebases characterized by structural degradation, resource inefficiency, and critical security vulnerabilities (Liu and Li, 2024).

Current evaluation paradigms rely heavily on binary metrics such as $pass@k$ or textual similarity measures like BLEU, which were originally designed for natural language translation rather than the complex, formal semantics of source code. These metrics offer an "execution-time success" that does not guarantee "maintenance-time sustainability". As the industry moves from viewing AI as an insignificant coding assistant to a primary code generator, the role of the software engineer is evolving from an author to a reviewer and auditor. Consequently, there is an urgent requirement to move beyond isolated functional assessment toward a

multidimensional quality governance framework (Evtikhiev *et al.* 2023).

1.1 Domain

The proposed research is situated within the domain of **AI-Assisted Software Quality Engineering (AI-SQE)**. This field examines the long-term impact of LLM-generated code on the health, security, and operational costs of software systems. The research specifically focuses on the formulation of a composite evaluation framework that addresses three critical pillars of software quality:

- **Long-term Maintainability and Software Evolution**: This sub-domain investigates the "add it and forget it" anti-pattern, where AI tools encourage the rapid addition of new code blocks rather than the refactoring of existing ones. This leads to an "Execution-Maintenance gap," characterized by an eightfold increase in code duplication and a 60% decline in refactoring activity compared to the pre-AI era (GitClear, 2025).

- **Security Posture and Vulnerability Resilience**: This area addresses the "Security Gap" inherent in LLM-generated outputs. Despite being functionally correct, AI-generated code frequently reproduces common flaws from its training data, such as the MITRE CWE Top 25[1].

- **Operational Efficiency and Sustainability**: This sub-domain explores the "Efficiency Paradox," where the time saved during development is offset by increased costs during operation. State-of-the-art LLMs often consumes more energy in complex algorithmic tasks like dynamic programming or bit manipulation.

1.2 Motivation

The motivation for this research is from optimizing models for functional success often results in a neglect of security best practices and architectural coherence. Ex-

[1] Highlights the most common and impactful software weaknesses

isting benchmarks are largely "snippet-level," failing to capture the complexity of real-world software where vulnerabilities and inefficiencies arise from interactions between multiple components and cross-file dependencies. Furthermore, current evaluation tools are "feature-rich but signal-poor"; while developers have access to data on duplication, bugs, and runtime, they lack a unified signal to guide the acceptance or rejection of AI-generated snippets based on overall codebase health (Lian *et al.* 2025).

Chapter 2

Problem Statement and Justification

2.1 The Research Gap

The current landscape of AI-assisted software development is dominated by a state of "Functional Myopia," where the evaluation of Large Language Models (LLMs) is primarily restricted to their ability to solve isolated, snippet-level programming tasks. While modern models demonstrate high proficiency on benchmarks such as **HumanEval** and **MBPP**, these assessments focus almost exclusively on functional correctness—whether the code performs its intended task—while ignoring the broader quality characteristics essential for production-grade software (Liu *et al.* 2023).

A critical disconnect exists between a model's capacity to produce code that "runs" in a controlled environment and its ability to generate codebases that are sustainable, secure, and resource-efficient in the long run. Modern literature suggests that real-world software engineering involves complex inter-dependencies, architectural constraints, and multi-file contexts that current snippet-level benchmarks fail to simulate. Consequently, there is a significant research gap regarding the formulation of a validated composite metric that can capture these multidimensional quality signals to ensure long-term codebase health.

2.2 Justification

The necessity of this research is underscored by three emerging paradoxes identified across the domains of efficiency, maintainability, and security.

2.2.1 The Efficiency Paradox: The Trade-off of Sustainability

While AI tools undoubtedly accelerate the code-writing process, this speed often comes at the cost of long-term operational sustainability. LLMs are fundamentally trained to prioritize producing a correct result over an efficient one, often leading to sub-optimal algorithmic choices. This creates a "runtime gap" where AI-generated solutions frequently consume significantly more memory, computing power, than human-expert solutions, particularly in complex areas such as dynamic programming or systems-level programming. For production software running at scale, the increased infrastructure costs and environmental impact of this "power-hungry" code can eventually outweigh the one-time productivity gains achieved during the development phase

2.2.2 The Success/Maintainability Paradox: The "Add it and Forget it" Anti-pattern

High functional scores on standard benchmarks often mask a structural degradation within the codebase, a phenomenon described as the "add it and forget it" anti-pattern. Longitudinal studies of global repositories indicate that the rise of AI-assisted coding has correlated with a sharp increase in code duplication and a simultaneous decline in refactoring activity. Because LLMs primarily operate through an autoregressive, next-token prediction mechanism within fixed context windows, they often struggle to recognize opportunities for code reuse across a larger system. Instead of modifying existing logic, AI assistants tend to suggest entirely new code blocks, leading to locally correct but globally redundant solutions that compound technical debt and increase the maintenance burden over

time.

2.2.3 The Security-Functionality Disconnect: The Alignment Tax

A critical finding in recent security research is that there is no direct correlation between a model's functional performance and its security posture. Models that excel at passing unit tests frequently generate "bare-bones" code that lacks essential defensive layers, such as input sanitization or proper credential management. This creates an "alignment tax" where models optimized for utility and correctness may inadvertently reproduce or even amplify known vulnerabilities found in their training data. Furthermore, enhanced "reasoning" capabilities in newer models do not inherently translate to safer code in repository-level settings, as the complexity of maintaining security constraints across multiple files and dependencies often overwhelms current model capacities (Sabra *et al.* 2025).

Chapter 3

Research Aims, Questions, and Objectives

3.1 Primary Aim

To develop and validate a **Composite Quality Index (CQI)** that mathematically unifies **Correctness**, **Maintainability**, **Security**, and **Efficiency** into a single governance framework for AI-assisted development.

3.2 Research Questions

Based on the critical gaps identified in current literature, this research seeks to address the following questions:

- **RQ1: The Metric Integration Problem** How can disparate quality dimensions, such as execution efficiency, structural maintainability, and security risk, be normalized and integrated into a single, mathematically sound composite metric?
- **RQ2: The Reasoning-Security Paradox** To what extent do enhanced reasoning capabilities in LLMs correlate with the production of more secure and maintainable codebases, and how does this "reasoning budget" affect the overall quality index?

- **RQ3: The Context-Efficiency Gap** How does the reliance on snippet-level generation rather than "whole-repository" reasoning contribute to structural degradation and resource inefficiencies in AI-generated code?
- **RQ4: Language-Specific Variability** How significantly do the inherent characteristics of programming languages influence the weighting of the final composite score?

3.3 Research Objectives

To answer the research questions and fulfill the primary aim, the following objectives have been established:

- **To quantify structural degradation[1] patterns** by analyzing the relationship between the increase in code duplication and the simultaneous decline in refactoring activity within AI-assisted repositories.
- **To identify novel technical debt categories** unique to LLM-integrated systems, such as "prompt debt," "hallucinated dependencies," and "architectural blindness," for inclusion in the maintainability assessment.
- **To evaluate the "Efficiency Paradox"** through empirical benchmarking of runtime performance, memory usage, and energy consumption across diverse programming languages using frameworks like EffiBench and Mercury.
- **To establish a "Security-Weighted" functional metric** that integrates traditional $pass@k$ success rates with vulnerability prevalence (VR) and severity scores (CVSS).
- **To formulate and validate a unified mathematical model (CQI)** that assigns dynamic weights to these four pillars, providing developers with a single, reliable signal for AI-generated codebase health.

[1] Long-term decline in the quality and organization of a software project's architecture

Chapter 4

Conclusion

Conclusion goes here.

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